Università di Pisa

Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

Candidato:
Alessandro Gori

Relatore:
Prof. Giacomo Mazza

Outline

Cuprates and Hubbard physics

Superconductivity in cuprates

Nobel Prize in Physics 1987

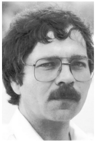


Photo from the Nobel Foundation archive.

J. Georg Bednorz

Prize share: 1/2

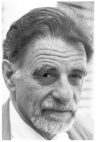


Photo from the Nobel Foundation archive.

K. Alexander Müller

Prize share: 1/2

Possible High T_c Superconductivity in the Ba – La – Cu – O System

J.G. Bednorz and K.A. Müller
IBM Zürich Research Laboratory, Rüschlikon, Switzerland

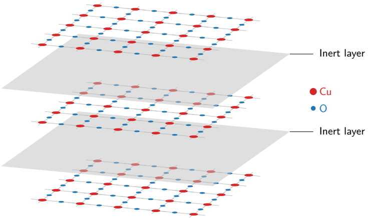
Received April 17, 1986

What makes cuprates interesting?

- Strong “unconventional” insulators;
- Rich phase diagram;
- Origin of Cooper pairing: still unclear;
- Highest critical temperatures for SC transition in this class of materials.

These materials are stackings of:

- 2D copper oxide active layers;
- intermediate layers of inert material.



Most of the cuprates physics is well described by the single-band Hubbard model on the square lattice:

$$\hat{H}_{\mathrm{HM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{On-site repulsion}} \qquad \text{with } t, U > 0.$$

The phase diagram of cuprates

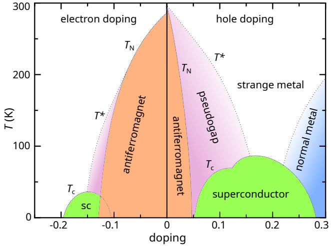
Cuprates exhibit various phases, among which:

- **Antiferromagnetic long-range order** at null and low doping;
- d-**wave superconductivity** as we inject or remove charge.

At **half-filling**, band theory would predict a metallic state. Instead we observe strong insulation.

$$\Downarrow$$

Cuprates are **not** band insulators.



Schematic phase diagram of cuprates. Author: Holger Motzkau, Wikimedia commons, CC BY-SA 3.0.

Mott localisation: a many-body insulator

Band theory assumption: e-e interactions are **negligible**.

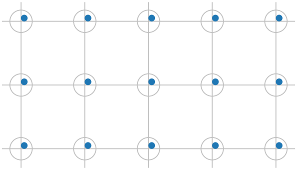
Strongly coupled Hubbard model:

$$U \gg t\,,$$

Big energetic cost of having two electrons on the same site.

$$\Downarrow$$

Insulation mechanism: electrons are **localised** as in a sort of "traffic jam".



Mott localisation: a many-body insulator

Band theory assumption: e-e interactions are **negligible**.

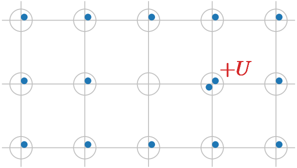
Strongly coupled Hubbard model:

$$U \gg t\,,$$

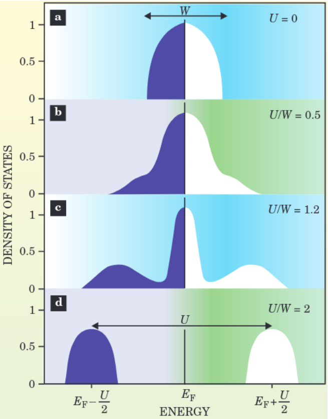
Big energetic cost of having two electrons on the same site.

$$\Downarrow$$

Insulation mechanism: electrons are **localised** as in a sort of "traffic jam".



Bands flattening: a signature of Mott localisation



Gabriel Kotliar and Dieter Vollhardt, 2004.
Physics Today 57 (March): 53–59.

(a) Free model, conventional DoS;

(b) Weak coupling, large coherent peak;

(c) Intermediate coupling, **narrow coherent peak**;

(d) Strong coupling, lower and upper Hubbard bands.

$$\Downarrow$$

Localisation = flattening of bare bands.

How does Mott localisation relate to antiferromagnetism and superconductivity?

Antiferromagnetism

Perturbation theory at strong coupling gives:

$$\hat{H}_{\mathrm{HM}} \simeq J \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \Delta E \,,$$

with antiferromagnetic exchange $J = 4t^2/U$.

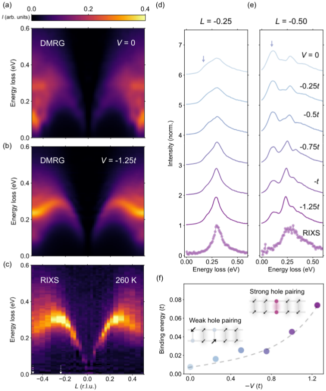
"Mottness": **the ground manifold is localised**.

Superconductivity

Phenomenology: we observe d-wave superconductivity, i.e. the superconducting gap is anisotropic under $\pi/2$ rotations $\mathcal{R}$,

$$\left\langle \hat{c}^\dagger_{\uparrow\mathbf{r}} \hat{c}^\dagger_{\downarrow\mathbf{r}+\Delta\mathbf{r}} \right\rangle = - \left\langle \hat{c}^\dagger_{\uparrow\mathcal{R}\mathbf{r}} \hat{c}^\dagger_{\downarrow\mathcal{R}(\mathbf{r}+\Delta\mathbf{r})} \right\rangle \,.$$

"Mottness": **non-local Cooper pairing**.



Hari Padma et al., 2025. **Physical Review X 15 (2): 021049**.

Incompatibilities with the spectrum of magnetic excitation in cuprates.

$$\Downarrow$$

Proposal: extension of the model with a **non-local attraction**,

$$\hat{H}_{\mathrm{EHM}} = \hat{H}_{\mathrm{HM}} \underbrace{-V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\hat{H}_V} .$$

$\hat{H}_V$ can act **as a source of d-wave superconductivity**.

The model (EHM):

$$\hat{H}_{\mathrm{EHM}} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\text{Local repulsion}} \underbrace{-V\sum_{\langle ij\rangle}\hat{n}_{i}\hat{n}_{j}}_{\text{Non-local attraction}} \qquad \text{with } t, U, V > 0.$$

The motivation: Recent experimental findings.

The idea: we know that the EHM can host d-wave superconductivity. Can we obtain Mott physics?

The method: theoretical and numerical Hartree-Fock technique.

The Hartree-Fock method

We want to approximate the true ground state $|\Psi_0\rangle$ of the EHM, with energy E_0.

Variational principle:

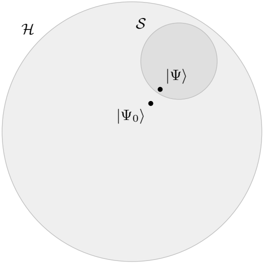
$$\forall \, |\Phi\rangle \in \mathcal{S} \subset \mathcal{H} \quad : \quad E[\Phi] \geq E_0 \, .$$

$$\Downarrow$$

Constrained minimisation procedure:

$$|\Psi\rangle \quad : \quad \left. \frac{\delta E[\Phi]}{\delta \, |\Phi\rangle} \right|_{|\Phi\rangle = |\Psi\rangle} = 0 \, .$$

The state $|\Psi\rangle$ is the best approximation to $|\Psi_0\rangle$ inside of $\mathcal{S}$.



Sketch of the Hilbert space $\mathcal{H}$ and a subset $\mathcal{S}$.

Hartree-Fock method: we identify $\mathcal{S}$ with the set of all the ground states of non-interacting hamiltonians,

$$\mathcal{S} = \left\{ |\Psi\rangle \;\middle|\; |\Psi\rangle \text{ is the ground-state of some } \hat{H}_{\mathrm{HF}} \right\} ,$$

where

$$\hat{H}_{\mathrm{HF}} \equiv \sum_{\alpha} E_\alpha \hat{\gamma}^\dagger_\alpha \hat{\gamma}_\alpha \quad \text{with} \quad \left\{ \hat{\gamma}_\alpha , \hat{\gamma}^\dagger_\beta \right\} = \delta_{\alpha\beta} ,$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.

$$\Downarrow$$

Wick's theorem:

$$\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle = \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\delta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\gamma \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}} .$$

The self-consistency mapping

Target parametric hamiltonian:

$$\hat{H}_{\mathrm{HF}}[\mathbf{v}] \equiv \sum_{\alpha} E_{\alpha}[\mathbf{v}] \hat{\gamma}^{\dagger}_{\alpha}[\mathbf{v}] \hat{\gamma}_{\alpha}[\mathbf{v}] \qquad \text{with ground state } |\Psi[\mathbf{v}]\rangle,$$

being $\mathbf{v}$ a vector of variational parameters. HF approximation:

$$|\Psi_0\rangle \simeq |\Psi[\mathbf{v}]\rangle \ ,$$

with $\mathbf{v}$ such that:

Variational picture

Energy is minimised,

$$\left. \frac{\delta E[\Phi]}{\delta |\Phi\rangle} \right|_{|\Phi\rangle = |\Psi[\mathbf{v}]\rangle} = 0 \, .$$

Self-consistency picture

A self-consistency problem is satisfied,

$$\mathrm{A}(\mathbf{v}) = \mathbf{v} \, ,$$

being A a non-linear map **to be determined**.

The iterative algorithm (iHF)

The HF solution is the **fixed point** of a non-linear map A,

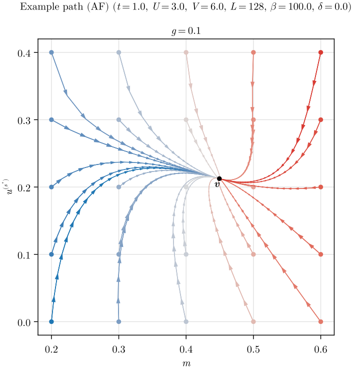
$$\mathrm{A}(\mathbf{v}) = \mathbf{v}\,.$$

$$\Downarrow$$

Strategy: "follow" the map up to the fixed point,

$$\mathbf{v}_{i+1} \equiv g\mathrm{A}(\mathbf{v}_i) + (1-g)\mathbf{v}_i\,.$$

with $i > 0$, $g \in [0,1]$.



Example path for the AF phase.

The HF method and symmetry breaking mechanisms

Symmetry breaking and HF method

Phase transition: spontaneous breaking of some symmetries of the model.

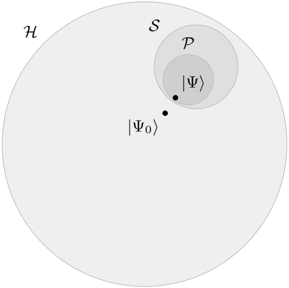
Phase X: shows symmetries Y, Z ...

$$\Downarrow$$

Further constraining of HF subspace to $\mathcal{P} \subset \mathcal{S}$, states with symmetries Y, Z ...

$$\Downarrow$$

Symmetry selection rules.



e.g.

$$\text{Charge is conserved} \quad \Longrightarrow \quad \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle = 0$$

The symmetries of the EHM

The model (EHM):

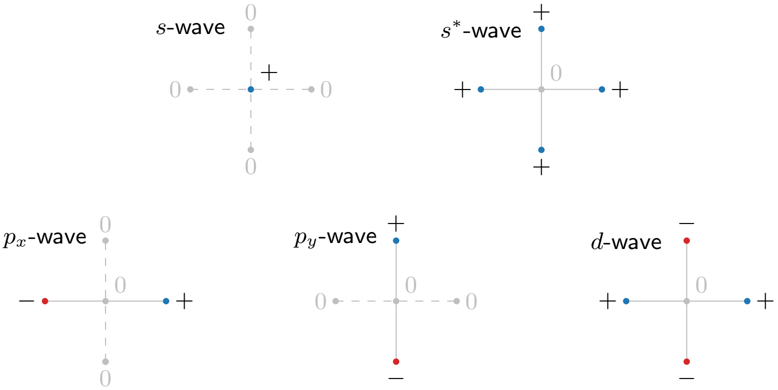
$$\hat{H}_{\mathrm{EHM}} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\text{Local repulsion}} \underbrace{-V\sum_{\langle ij\rangle}\hat{n}_{i}\hat{n}_{j}}_{\text{Non-local attraction}} \qquad \text{with } t, U, V > 0,$$

and its symmetries:

- *Translational invariance* in directions x and y, $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$.
  If broken: Non-uniform spin density.
- *Rotational invariance* under $\pi/2$ rotations, C_4.
  If broken: nematicity.
- *Gauge invariance* (charge conservation), $\mathrm{U}(1)$.
  If broken: Superconductivity.
- *Global spin rotational invariance*, $\mathrm{SU}(2)$.
  If broken: Magnetism.

The spatial symmetry channels

It is useful to introduce five symmetry structures:



The normal phase

The self-consistency problem for the normal phase

Normal phase: no broken symmetries. From the Fock channel,

$$\tilde{t}_{ij\sigma} \equiv \begin{cases} t - V\langle \hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\rangle & \text{if } i, j \text{ are NNs,} \\ 0 & \text{otherwise.} \end{cases}$$

$$\Downarrow$$

Reciprocal space: let $u_{\mathbf{k}} \equiv \langle \hat{n}_{\mathbf{k}} \rangle$ be the occupation and $u^{(\gamma)}$ its harmonics,

$$\tilde{t}^{(s^*)} = t - \frac{u^{(s^*)}V}{2}\,.$$

Bare bands:

$$\epsilon_{\mathbf{k}} = -2t(\cos k_x + \cos k_y)$$

Bands renormalisation effect:

$$\tilde{\epsilon}_{\mathbf{k}} = \epsilon_{\mathbf{k}} + u^{(s^*)} V(\cos k_x + \cos k_y)$$

That’s all, thank you!

Backup slides